

1. AWT Event Handling – Message Exchange

Server.java

```
import java.awt.*;
import java.awt.event.*;
import java.net.*;
import java.io.*;

public class Server extends Frame implements ActionListener {
    TextField text;
    Button send;
    ServerSocket ss;
    Socket socket;
    DataOutputStream out;

    Server() {
        setTitle("Server");
        setSize(400, 150);
        setLayout(new FlowLayout());

        text = new TextField(25);
        send = new Button("Send");

        add(text);
        add(send);

        send.addActionListener(this);

        setVisible(true);

        try {
            ss = new ServerSocket(5000);
            socket = ss.accept();
            out = new DataOutputStream(socket.getOutputStream());
        } catch (Exception e) {
            System.out.println(e);
        }

        addWindowListener(new WindowAdapter() {
            public void windowClosing(WindowEvent e) {
                System.exit(0);
            }
        });
    }

    public void actionPerformed(ActionEvent e) {
        try {
            String msg = text.getText();

            System.out.println("Server: " + msg);

            out.writeUTF(msg);

            text.setText("");
        }
    }
}
```

```

        } catch (Exception ex) {
            System.out.println(ex);
        }
    }

    public static void main(String[] args) {
        new Server();
    }
}

```

Client.java

```

import java.awt.*;
import java.awt.event.*;
import java.net.*;
import java.io.*;

public class Client extends Frame implements ActionListener {
    TextField text;
    Button send;
    Socket socket;
    DataOutputStream out;

    Client() {
        setTitle("Client");
        setSize(400, 150);
        setLayout(new FlowLayout());

        text = new TextField(25);
        send = new Button("Send");

        add(text);
        add(send);

        send.addActionListener(this);

        try {
            socket = new Socket("localhost", 5000);
            out = new DataOutputStream(socket.getOutputStream());
        } catch (Exception e) {
            System.out.println(e);
        }

        setVisible(true);

        addWindowListener(new WindowAdapter() {
            public void windowClosing(WindowEvent e) {
                System.exit(0);
            }
        });
    }

    public void actionPerformed(ActionEvent e) {
        try {
            String msg = text.getText();

```

```

        System.out.println("Client: " + msg);

        out.writeUTF(msg);

        text.setText("");
    } catch (Exception ex) {
        System.out.println(ex);
    }
}

public static void main(String[] args) {
    new Client();
}
}

```

2. Java Login Screen

```

import java.awt.*;
import java.awt.event.*;

public class Login extends Frame implements ActionListener {
    TextField user;
    TextField password;
    Button login;
    Label result;

    Login() {
        setTitle("Login");
        setSize(400, 250);
        setLayout(new GridLayout(4, 2));

        Label l1 = new Label("Username:");
        Label l2 = new Label("Password:");

        user = new TextField();
        password = new TextField();
        password.setEchoChar('*');

        login = new Button("Login");
        result = new Label("");

        add(l1);
        add(user);

        add(l2);
        add(password);

        add(login);
        add(result);

        login.addActionListener(this);

        addWindowListener(new WindowAdapter() {
            public void windowClosing(WindowEvent e) {
                System.exit(0);
            }
        });
    }
}

```

```

        }
    });

    setVisible(true);
}

public void actionPerformed(ActionEvent e) {
    String username = user.getText();
    String pass = password.getText();

    System.out.println("Username: " + username);
    System.out.println("Password: " + pass);

    if (username.length() > 0 && pass.length() > 0)
        result.setText("Login Successful");
    else
        result.setText("Enter details");
}

public static void main(String[] args) {
    new Login();
}
}

```

3. Basic Calculator Using AWT

```

import java.awt.*;
import java.awt.event.*;

public class Calculator extends Frame implements ActionListener {
    TextField n1, n2, result;
    Button add, sub, mul, div;

    Calculator() {
        setTitle("Calculator");
        setSize(400, 250);
        setLayout(new GridLayout(5, 2));

        n1 = new TextField();
        n2 = new TextField();
        result = new TextField();
        result.setEditable(false);

        add = new Button("+");
        sub = new Button("-");
        mul = new Button("*");
        div = new Button("/");

        add(new Label("First Number"));
        add(n1);

        add(new Label("Second Number"));
        add(n2);

        add(new Label("Result"));
    }
}

```

```

        add(result);

        add(add);
        add(sub);
        add(mul);
        add(div);

        add.addActionListener(this);
        sub.addActionListener(this);
        mul.addActionListener(this);
        div.addActionListener(this);

        addWindowListener(new WindowAdapter() {
            public void windowClosing(WindowEvent e) {
                System.exit(0);
            }
        });

        setVisible(true);
    }

    public void actionPerformed(ActionEvent e) {
        double a = Double.parseDouble(n1.getText());
        double b = Double.parseDouble(n2.getText());
        double ans = 0;

        if (e.getSource() == add)
            ans = a + b;
        else if (e.getSource() == sub)
            ans = a - b;
        else if (e.getSource() == mul)
            ans = a * b;
        else if (e.getSource() == div)
            ans = a / b;

        result.setText("" + ans);

        System.out.println("First Number: " + a);
        System.out.println("Second Number: " + b);
        System.out.println("Result: " + ans);
    }

    public static void main(String[] args) {
        new Calculator();
    }
}

```

4. Print Input on Console and Save Input to File

```

import java.awt.*;
import java.awt.event.*;
import java.io.*;

public class InputFile extends Frame implements ActionListener {
    TextField text;

```

```

Button save;

InputFile() {
    setTitle("Input");
    setSize(400, 150);
    setLayout(new FlowLayout());

    text = new TextField(25);
    save = new Button("Save");

    add(text);
    add(save);

    save.addActionListener(this);

    addWindowListener(new WindowAdapter() {
        public void windowClosing(WindowEvent e) {
            System.exit(0);
        }
    });

    setVisible(true);
}

public void actionPerformed(ActionEvent e) {
    String input = text.getText();

    System.out.println("Input: " + input);

    try {
        FileWriter fw = new FileWriter("input.txt", true);
        fw.write(input + "\n");
        fw.close();

        System.out.println("Input saved to file");
    } catch (Exception ex) {
        System.out.println(ex);
    }

    text.setText("");
}

public static void main(String[] args) {
    new InputFile();
}
}

```